The University Of Sheffield.

Ancillary Material: Computer Answer Sheets

DEPARTMENT OF COMPUTER SCIENCE Autumn Semester 2021-22

TEXT PROCESSING 2 hours

Answer ALL questions.

All questions carry equal weight. Figures in square brackets indicate the percentage of available marks allocated to each part of a question.

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1. a) Unicode code points are mapped onto encoding forms as shown in the following table.

Scalar Value	UTF-16	UTF-8 Bit Distribution			
		1st Byte	2nd Byte	3rd Byte	4th Byte
000000000xxxxxxx	000000000xxxxxxx	0xxxxxxx			
00000yyyyyxxxxxx	00000yyyyyxxxxxx	110yyyyy	10xxxxxx		
zzzzyyyyyyxxxxxx	zzzzyyyyyyxxxxxx	1110zzzz	10yyyyyy	10xxxxxx	
uuuuuzzzzyyyyyyxxxxxx	110110wwwwzzzzyy+ 110111yyyyxxxxxx	11110uuu[a]	10uuzzzz	10yyyyyy	10xxxxxx

[a] Where uuuuu = wwww + 1

The Chinese character 柬 has Unicode code point U+67EC. What are the UTF-16 and UTF-8 encoding forms for 柬? Please express your answer in hexadecimal. Show how you derived your answer. [20 marks]

b) We want to compress a large corpus of text of the (fictitious) language *Fontele*. The writing script of Fontele employs only the six letters found in the language name (f,o,n,t,e,l) and the symbol ⊔, used as a 'space' between words. Corpus analysis shows that the probabilities of these seven characters are as follows:

Symbol	Probability
e	0.3
f	0.04
l	0.26
n	0.2
t	0.04
o	0.1
⊔	0.06

(i) Show how to construct a Huffman code tree for Fontele, given the above probabilities for its characters. Use your code tree to assign a binary code for each character. [20%]

(ii) Use your code to encode the message "telefone⊔noel" and show the resulting binary representation. For this string, how much does your encoding save over a minimal fixed length binary character encoding for a 7 character alphabet? [10%]

(iii) Show how a *canonical* Huffman code can be derived from the Huffman code that you created for Fontele in 1(b)(i). What are the advantages of using a canonical Huffman code? [20%]

QUESTION CONTINUED ON THE NEXT PAGE

c) LZ77 is a popular compression method, used in common compression utilities such as *gzip*. The following shows some possible LZ77 encoder output, assuming the encoding representation presented in the lectures of the Text Processing module:

$$\langle 0,0,b\rangle\langle 0,0,a\rangle\langle 0,0,d\rangle\langle 3,3,b\rangle\langle 1,2,a\rangle\langle 1,2,d\rangle\langle 1,2,a\rangle\langle 9,2,a\rangle$$

(i) Explain the encoding representation. [10%]

(ii) What output would be produced by decoding the above representation? Show how you derived your answer. [20%]

2. Consider the following documents and query:

Document 1: They sailed to the port for a good dinner and sailed home after.

Document 2: Ruby port is very good after a meal, any meal.

Query: good ports after dinner

a) Show how Documents 1 and 2 below would be stored in an *inverted index*, assuming you have applied the term manipulation methods of capitalisation, punctuation removal, stop word removal and stemming to them and assuming the stoplist: {a, and, any, for, is, the, they, to} [20%]

b) Compute the similarity between the query and each document using the cosine metric, using term frequency values for the term weights in the document vectors. Which document is most similar to the query? [30%]

c) Describe and explain the TF.IDF (*term frequency inverse document frequency*) term weighting scheme, including the formula (or formulae) for computing these values as part of your answer. Discuss how use of this term weighting scheme (rather than just term frequency) affects the outcome of the ranking calculation made in 2(b), showing how you arrive at your answer. [20%]

QUESTION CONTINUED ON THE NEXT PAGE

d) Two ranked retrieval systems – System 1 and System 2 – each return 10 documents for a given query. The following table shows, for each system, whether the document returned at each rank is relevant ($\checkmark$) or not ($\times$). It is known that there are 8 relevant documents for this query, within an overall collection of 100 documents.

Rank	System 1	System 2
1	$\times$	$\times$
2	$\checkmark$	$\checkmark$
3	$\times$	$\times$
4	$\times$	$\checkmark$
5	$\checkmark$	$\times$
6	$\checkmark$	$\times$
7	$\times$	$\times$
8	$\times$	$\checkmark$
9	$\times$	$\times$
10	$\checkmark$	$\checkmark$

(i) Explain the Precision, Recall and F-measure metrics, as used to evaluate information retrieval systems. Compute the performance of these two systems in terms of these three metrics. [15%]

(ii) Given that the two systems return the same number of relevant documents, suggest an alternative metric for evaluating their performance, and apply it to determine which system does better. [15%]

3. a) List two advantages of lexicon-based Sentiment Analysis approaches and two disadvantages. Identify which is the main disadvantage of binary approaches that can be solved using graded approaches. [5%]

b) Given sentences (1) and (2) below:

1. *I am 5 foot 6 and a little curvy and I got a size 10.*
2. *The material on the top looks and feels very cheap that even just pulling on it will cause it to rip the fabric.*

Considering the Sentiment Analysis task, would both sentences be treated in the same way? Define which sentence is useful for Sentiment Analysis and explain why. [10%]

c) Specify the quintuple of Bing Liu's model for Sentiment Analysis, identifying all the components in the example below. Identify all the features present in the text and, for each of them, indicate the sentiment value as positive or negative.

"Last night I went to a small restaurant on South Road. I have passed by this road several times and I have never noticed this small place. The food is excellent and the staff are very friendly. The portions are huge, so it is great value for money. I had a grilled tuna and it was fantastic!! The only drawback is that the restaurant is very small (depending on the place you are seated you need to move several times to give space for other clients)."

Billy S. posted on 15th March 2015. [15%]

d) Explain the graded (gradable) lexicon-based approach for Sentiment Analysis, including the six rules presented during our lectures. Given the following sentences and sentiment lexicon, apply this approach to classify each sentence in S1 – S4 as positive, negative or neutral. Show the final emotion score for each sentence and also how this score was devised. Give any general rules that you used to calculate this score as part of your answer. [30%]

Lexicon	
amazing	3
boring	-2
cold	-1
good	2
sad	-2
Intensifiers and Emoticons	
extremely	3
☹	-2

(S1) This product is amazing, but it costs $500 ☹
(S2) The new IPhone 5 is AMAZING!
(S3) The food was boring and extremely cold.
(S4) I am feeling good today, despite being sad about losing my phone.

QUESTION CONTINUED ON THE NEXT PAGE

e) (i) Explain how a Naive Bayes classifier can be trained and then used to predict the polarity class (positive or negative) of a subjective text. Be sure to give the mathematical formulation of the Naive Bayes classifier and explain its main components. [15%]

(ii) Suppose you are given the following set of labelled examples as training data:

Doc	Words	Class
1	A <u>sensitive</u>, <u>moving</u>, <u>brilliant</u> work	Positive
2	An <u>edgy</u> thriller that delivers a <u>surprising</u> punch	Positive
3	A <u>sensitive</u>, <u>insightful</u>, <u>flamboyant</u> film	Positive
4	Neither <u>revelatory</u> nor truly <u>edgy</u> – merely crassly <u>flamboyant</u> and comedically <u>labored</u>	Negative
5	<u>Unlikable</u>, <u>uninteresting</u>, <u>unfunny</u>, and completely, utterly <u>inept</u>	Negative
6	A sometimes <u>incisive</u> and <u>sensitive</u> portrait that is undercut by its <u>awkward</u> structure and . . .	Negative
7	It' s a sometimes <u>interesting</u> remake that doesn' t compare to the <u>brilliant</u> original	Negative

Using as features just the adjectives (underlined words), how would a Naive Bayes sentiment analyser trained on these examples classify the sentiment of the new, unseen text shown below?

Doc	Words	Class
8	A <u>flamboyant</u> romcom, <u>sensitive</u> but <u>awkward</u> at times.	???

You should assume that the texts are tokenised, lowercased and that punctuation was removed. You DO NOT need to smooth feature counts. (Tip: you do not need to calculate the likelihood for all features). [25%]

4. a) Consider the following sentence:

William MacDonald took the bus from Kelham Island to Manor Park, aiming to visit the Sheffield Manor Lodge.

Annotate the sentence using the labelling that the BIO approach adopts to labelling named entities for training a named entity recogniser, assuming the three entity types: person, organisation and location. [25%]

b) What is the difference between the tasks of Information Retrieval and Information Extraction. Cite one strength and one weakness for each task. [20%]

c) Explain the Entity Linking task. Why is it a difficult task? [20%]

d) (i) Name the four different approaches for Information Extraction discussed in our lectures. [15%]

(ii) Which type of approach is DIPRE (Dual Iterative Pattern Relation Expansion)? Define the five steps in DIPRE. [20%]

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